

# Domain-Adapted Llama 3 8B

Software engineering Q&A with QLoRA

30.42%

LOWER HELD-OUT  
ANSWER PERPLEXITY

Base 7.9560 -> Tuned 5.5359 | 100 paired examples | 18,798 answer tokens

## What this project demonstrates

An end-to-end adaptation workflow that converts a large public preference corpus into a controlled supervised fine-tuning experiment, trains a parameter-efficient adapter on an A100, and compares the adapted and unadapted model under identical inference conditions.

50,500

real-world Q&A examples cached

0.26%

of parameters trained with LoRA

2:44:33

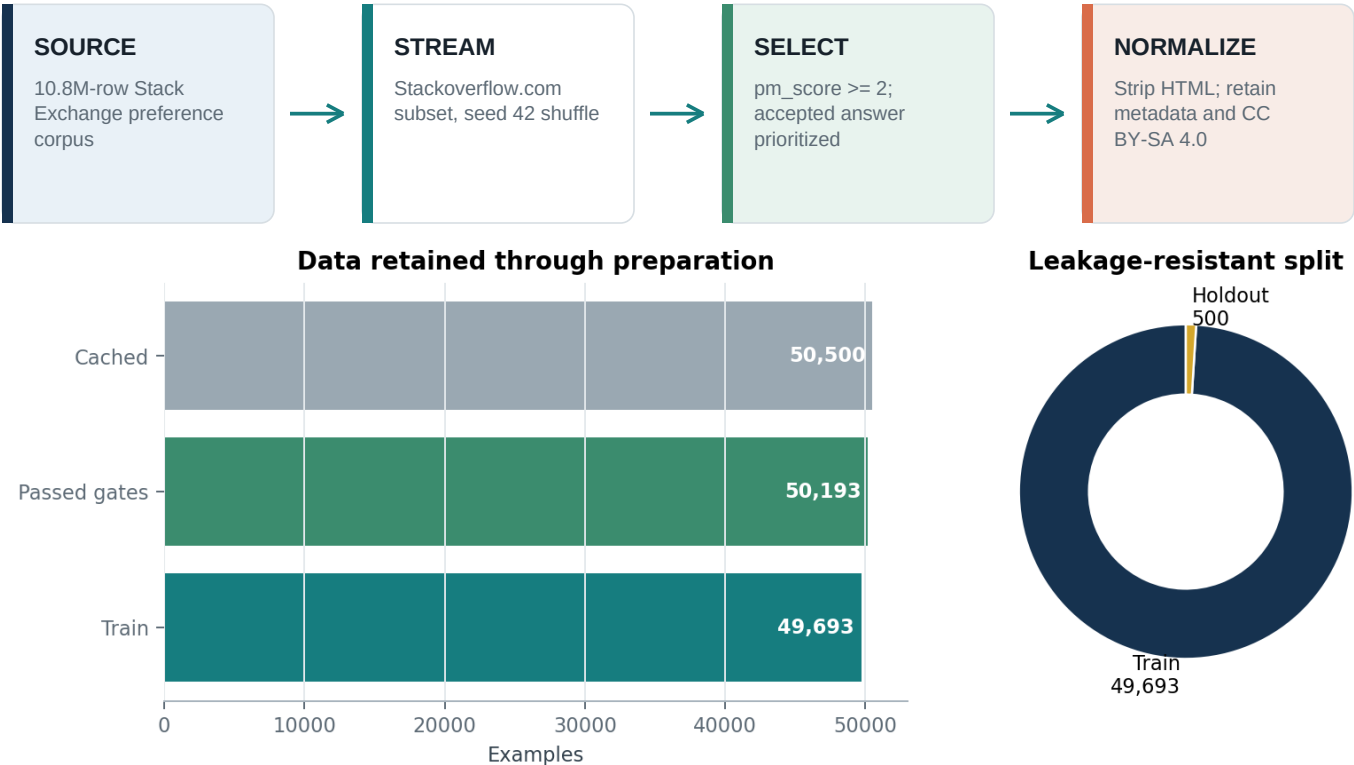
single-A100 training runtime

### PORTFOLIO SIGNAL

The result is evidence that a small trainable adapter can materially shift an 8B model toward the target answer distribution while preserving a frozen quantized backbone. It demonstrates data engineering, efficient training, experimental control, statistical evaluation, and deployment-oriented artifact design.

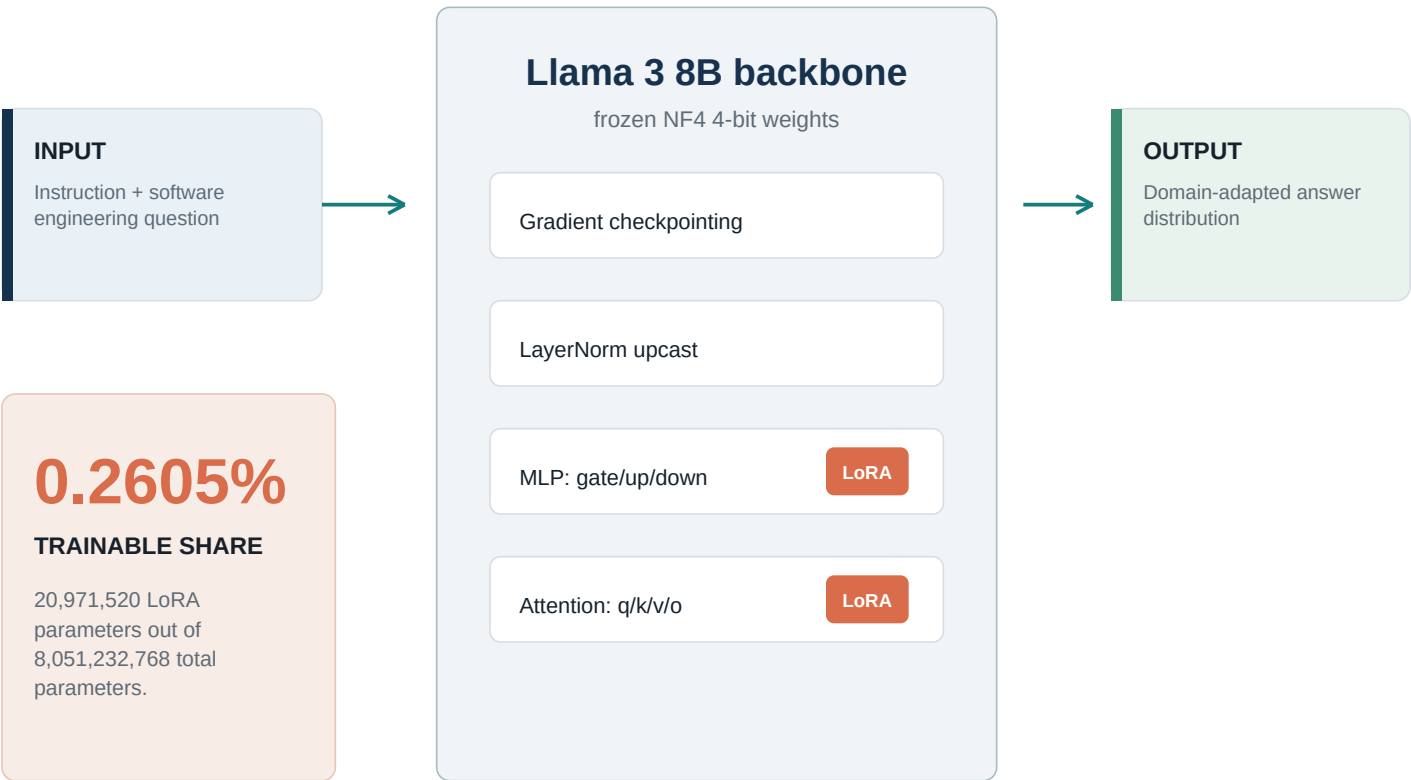
# From public Q&A to controlled SFT data

A deterministic pipeline selects stronger Stack Overflow answers, normalizes text, enforces quality gates, and isolates a fixed holdout before training.



# Efficient adaptation, not full retraining

QLoRA keeps the Llama 3 8B backbone frozen in 4-bit form while training low-rank updates on every linear projection.



## Training configuration

**BASE MODEL**

Meta-Llama-3-8B-Instruct

**METHOD**

LoRA on all linear targets

**QUANTIZATION**

4-bit NF4 + double quantization

**PRECISION**

bfloat16 compute; LayerNorm FP32

**SEQUENCE**

2,048 tokens; packing enabled

**BATCH**

2 x 8 accumulation = 16 effective

**SCHEDULE**

1 epoch; cosine; LR 2e-4

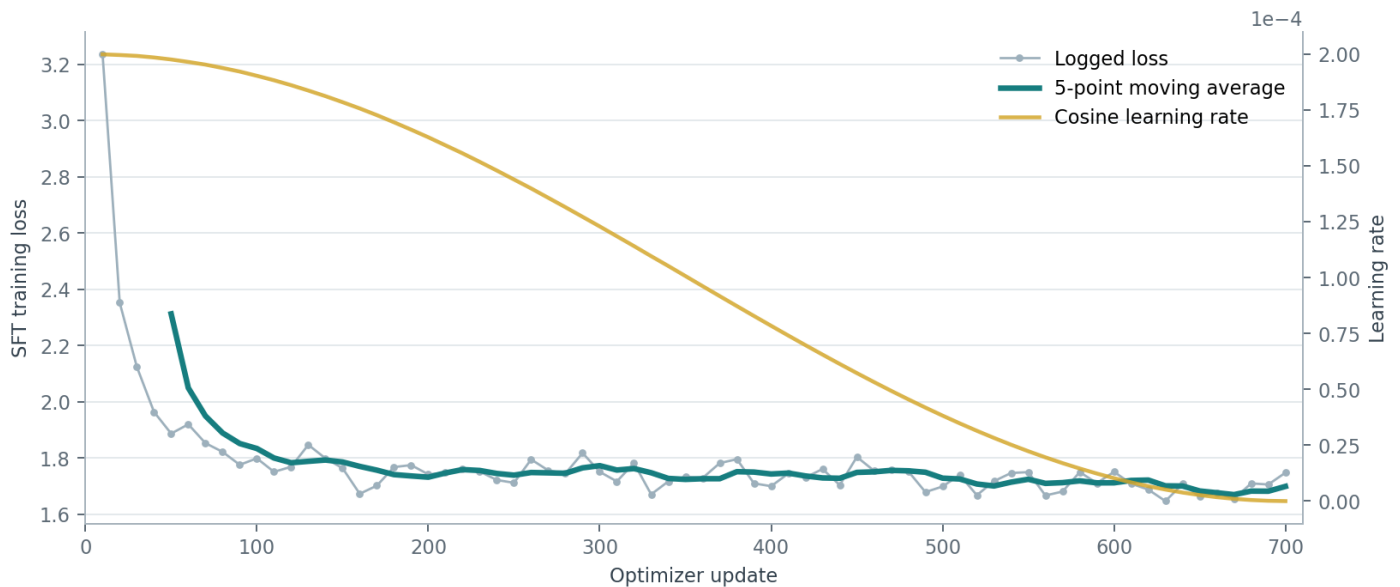
**CHECKPOINTS**

every 50 steps; retain 2

Engineering meaning: the adapter is portable and inexpensive to store, while disabling it recovers the exact quantized baseline for a controlled A/B comparison.

# Stable one-epoch adaptation on a single A100

The curve shows rapid early domain adaptation followed by a stable plateau under cosine decay.

**701**

optimizer updates

**2:44:33**

wall-clock runtime

**1.7811**

reported train loss

**1.135**

packed sequences/sec

## OBSERVED DYNAMICS

Mean loss fell from 2.0732 across the first 10 logs to 1.6911 across the final 10, an 18.43% reduction. Gradient norms stabilized after the initial updates.

## WHAT THE CURVE CANNOT PROVE

Training loss alone does not establish better answers. No in-training eval loss was recorded, so the claim depends on the separate frozen holdout evaluation shown next. This separation is deliberate and avoids tuning to the test metric.

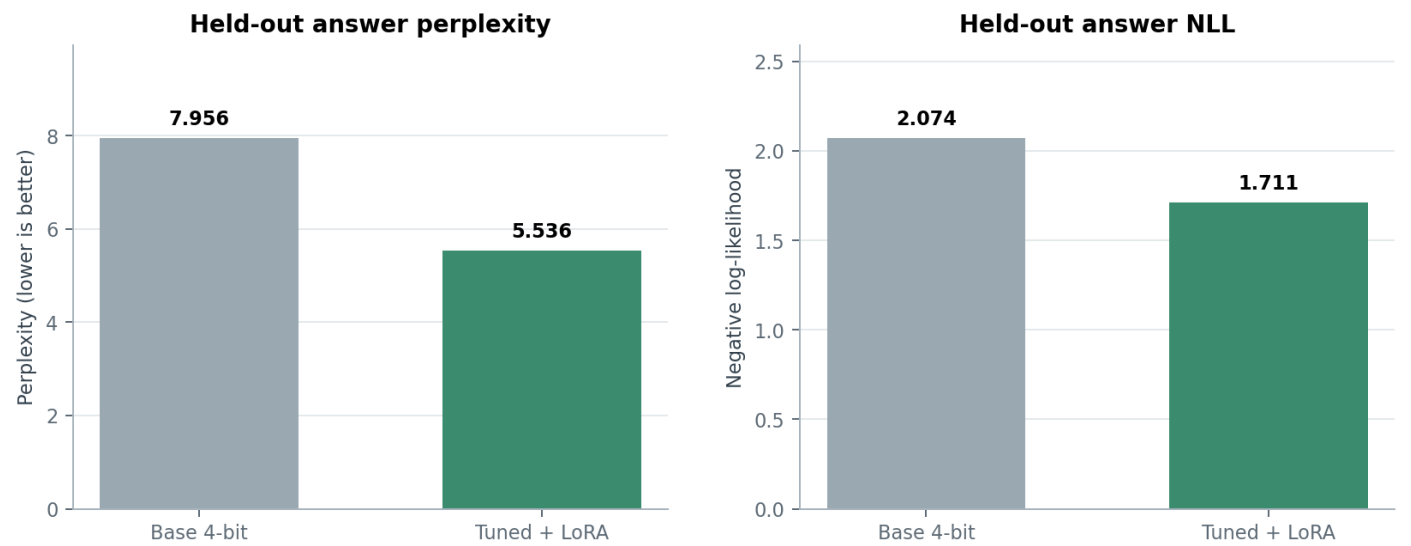
# A controlled paired comparison

The experiment changes one factor only: whether the trained LoRA adapter is active.



# The adapter materially improves held-out fit

All displayed values come from the notebook's saved evaluation output.



30.42%

PERPLEXITY REDUCTION

7.9560 base -> 5.5359 tuned

100%

PAIRED EXAMPLE WIN RATE

100 examples; 18,798 answer tokens

CONFIDENCE CHECK

The 95% bootstrap interval for unweighted mean paired NLL improvement is [0.475145, 0.590001]. Because the entire interval is above zero, the direction of improvement is consistent across resampled evaluation sets.

DEFENSIBLE CLAIM

On a fixed 100-example holdout subset, QLoRA tuning reduced answer perplexity by 30.42% versus the same 4-bit base model.

DO NOT OVERCLAIM

Perplexity is not factual accuracy or production quality. The result shows stronger fit to held-out reference answers; human and execution-based evaluation remain necessary.

# What the project proves - and what comes next

A credible engineering report separates measured evidence from the next validation steps.

### PROVEN IN THIS RUN

- Reproducible ingestion and atomic caching of 50,500 real-world records.
- Quality gates, deterministic split, and LLaMA-Factory registration.
- Single-A100 QLoRA training with 0.2605% trainable parameters.
- Controlled adapter-on versus adapter-off evaluation on held-out answers.
- 30.42% lower perplexity with a paired bootstrap interval above zero.

### LIMITATIONS

- Final evaluation covers 100 of the 500 held-out examples.
- Reference likelihood does not directly score correctness, safety, or usefulness.
- Community votes and acceptance are imperfect proxies for answer quality.
- No independent contamination audit or executable-code benchmark was run.
- The adapter and evaluation artifacts live in ephemeral Colab storage unless exported.

### NEXT EXPERIMENTS

- Run all 500 holdout examples and stratify by topic and answer length.
- Add human pairwise review for correctness, relevance, and code validity.
- Execute code-bearing answers in sandboxes and record pass rates.
- Benchmark latency, memory, and throughput before and after adapter merge.
- Publish the adapter, model card, data statement, and reproducibility manifest.

## Evidence and source links

|                                     |                                                                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project notebook                    | <a href="#">/Users/wuhongda/Documents/jobs/ai_project_artifacts/01_llm_fine_tuning_pipeline.ipynb</a>                                                           |
| Stack Exchange preference dataset   | <a href="https://huggingface.co/datasets/HuggingFaceH4/stack-exchange-preferences">https://huggingface.co/datasets/HuggingFaceH4/stack-exchange-preferences</a> |
| Meta Llama 3 8B Instruct model card | <a href="https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct">https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct</a>                             |
| LLaMA-Factory                       | <a href="https://github.com/hiyouga/LlamaFactory">https://github.com/hiyouga/LlamaFactory</a>                                                                   |
| QLoRA paper                         | <a href="https://arxiv.org/abs/2305.14314">https://arxiv.org/abs/2305.14314</a>                                                                                 |
| PEFT adapter controls               | <a href="https://huggingface.co/docs/peft/main/package_reference/peft_model">https://huggingface.co/docs/peft/main/package_reference/peft_model</a>             |
| Perplexity definition               | <a href="https://huggingface.co/docs/transformers/v4.33.0/en/perplexity">https://huggingface.co/docs/transformers/v4.33.0/en/perplexity</a>                     |

### REPORTING RECOMMENDATION

Use the 30.42% figure with its metric and sample size: 'reduced held-out answer perplexity by 30.42% on a 100-example paired evaluation.' Replace it after the full 500-example run rather than presenting perplexity as a generic quality score.